THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · EASY LEVEL CHAPTER TEST SET 3

Chapter 13: **Earth as a System: Energy, Matter and Life** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Switching off lights saves: 1

(a) air **(b)** soil **(c)** energy **(d)** water

2. The ozone layer lies in the: 1

(a) crust **(b)** stratosphere **(c)** troposphere **(d)** core

3. Plants take in CO2 for: 1

(a) photosynthesis **(b)** digestion **(c)** respiration **(d)** excretion

4. Rainwater harvesting saves: 1

(a) coal **(b)** water **(c)** oil **(d)** air

5. CFCs damage the: 1

(a) forest **(b)** soil **(c)** oceans **(d)** ozone layer

6. Nitrogen in air is about: 1

(a) 1 per cent **(b)** 21 per cent **(c)** 78 per cent **(d)** 50 per cent

7. Plastic bags should be: 1

(a) dumped **(b)** thrown in rivers **(c)** burnt **(d)** avoided

8. Global warming is the rise in the Earth's: 1

(a) mass **(b)** size **(c)** temperature **(d)** speed

9. Nitrogen-fixing bacteria live in the root nodules of: 1

(a) wheat **(b)** pulses **(c)** maize **(d)** rice

10. Speed of light is about: 1

(a) 30 m/s **(b)** 3 x 10^8 m/s **(c)** 3 x 10^5 m/s **(d)** 340 m/s

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. What is nitrogen fixation? 2

12. What is the ozone layer? 2

13. What is transpiration? 2

14. What is Mission LiFE? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. State three processes that add CO2 to the air. 3

16. State three features of the equator and the poles. 3

17. State three components of the hydrosphere. 3

OR

State three ways to reduce global warming. 3

18. State three roles of bacteria in the nitrogen cycle. 3

OR

State three components of the geosphere. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain the carbon cycle. 5

OR

Explain the importance of solar energy. 5

20. Explain the nitrogen cycle. 5

OR

Explain the role of forests in the Earth system. 5

END OF QUESTION PAPER — Total: 40 Marks

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